

Rupesh Devasam

MD, USA | +1 667 419 0966 | rupeshdevasam@gmail.com | <https://www.linkedin.com/in/rupeshdevasam/> | <https://github.com/Rupeshdevasam>

Portfolio: <https://rupeshdevasam.vercel.app/>

SUMMARY

I'm a Computer Science graduate student at UMBC with experience in AI and full-stack development, specializing in React, Vue.js, FastAPI, Node.js, and Python. I enjoy building scalable web applications, optimizing performance, and creating efficient APIs. Alongside full-stack development, I have hands-on experience in AI and machine learning projects, including federated learning, privacy-preserving model training, and neural network development. I'm seeking **Software Engineering Internship or Full-Time Roles** where I can apply both my full-stack and AI skills to build impactful, innovative solutions.

EDUCATION

University of Maryland Baltimore County, Baltimore, Maryland, USA

Aug 2024 - May 2026

Master of Science in Computer Science

Jawaharlal Nehru Technological University Kakinada (AITAM), Andhra Pradesh, INDIA

Aug 2016 - Sep 2020

Bachelor of Science in Computer Science

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++, C, PHP, Shell scripting

Frontend technologies: React.js, Vue.js, Tailwind, HTML, CSS, SASS, Three.js, RWD

Backend technologies: Node.js, Express.js, Flask, FastAPI

Database technologies: MongoDB, PostgreSQL, MySQL, Redis, SQL

Software & Tools: Git, Postman, Android Studio, Atlassian tools (Jira, Confluence), Figma

Frameworks & Libraries: MERN, REST, Pandas, NumPy, Matplotlib, Seaborn

Cloud Platforms: AWS (Lambda, S3, EC2, DB2), Azure AI, Azure Web apps

Operating Systems: Windows, Linux, MacOS

Other Skills: Docker, Data Preprocessing, NLP, Computer Vision, LLMs, LLM-Pretraining, Computer networks, Machine learning algorithms, Neural networks, Federated learning, Full-Stack Development, Web Development, API Development, Application Development, Process Automation, UI/UX Design, Data Visualization, CI/CD Pipelines, Mobile application development

EMPLOYMENT EXPERIENCE

SDE Intern, Centennial Technologies, Virginia, USA

Jun 2025 – Aug 2025

- Recognized as **Sprint Champion** for consistently delivering high-impact features and ensuring smooth full-stack integration within tight deadlines
- Spearheaded development of **CHAAP**, an FDA regulatory support application, built **from scratch** using **React.js (TypeScript)**, **Python Flask**, and **Azure AI services**.
- Contributed across the **full stack**: designing frontend UI/UX, building RESTful backend APIs, and integrating **Azure AI models** for document analysis and compliance automation.
- Collaborated in setting up the complete architecture, including **frontend-backend integration**, model API pipelines, authentication flows, and deploying applications on **Azure Web Apps**.
- Focused on building scalable, maintainable components with **React**, ensuring smooth regulatory workflows and intuitive user experience.
- Worked in an Agile environment, contributing to sprint planning, peer code reviews, and regular product demos.

Software Engineer II, Survey Monkey, Bangalore, India

Mar 2024 – Jun 2024

- Developed and enhanced full-stack features for high-traffic survey applications using **React**, **FastAPI**, and **Python**, improving performance and scalability for **100K+ users**.
- Built reusable UI components in **React** and ensured seamless integration with **FastAPI** backend, reducing development time for new features by 30%.
- Designed and optimized **RESTful APIs** in **FastAPI**, improving data retrieval efficiency and reducing API response times by 25%.
- Collaborated with cross-functional teams using **Figma** to refine UI/UX designs, enhancing user experience and accessibility across web platforms.
- Actively participated in **daily code concaves**, contributing to architectural discussions, code reviews, and best practices to improve code quality and maintainability.

Web Developer, Visnet AI, Hyderabad, India

Apr 2023 – Feb 2024

- Led the development and implementation of web strategies using **Vue.js**, **FastAPI**, and **MongoDB**, enhancing user

experience and engagement across multiple platforms.

- Designed and optimized **RESTful APIs** in **FastAPI**, improving data retrieval efficiency and reducing response times by 30%.
- Built and maintained scalable frontend components in **Vue.js**, reducing development time for new features by 35% and improving UI consistency.
- Analyzed **web performance metrics** and implemented optimizations, leading to a **25% improvement in page load speed**.
- Collaborated with cross-functional teams to align web initiatives with business goals, driving brand visibility and platform scalability.

Full stack developer & Intern, Valuelabs, Hyderabad, India

Aug 2019 – Mar 2023

- Developed and maintained scalable web applications using **React, Vue.js, Node.js, and Express**, improving frontend performance and user experience for enterprise clients.
- Designed and improved **RESTful APIs** in **Node.js and Express**, reducing response times by 30% and enhancing data retrieval efficiency.
- Integrated SQL and MongoDB databases, optimizing queries and improving data processing speed by 25%.

SDE Intern, Tata Consultancy Services, Hyderabad, India

May 2019 – July 2019

- Developed and enhanced Flask-based web applications, improving backend efficiency and reducing response times by 25%.
- Designed and implemented dynamic frontend interfaces using HTML, CSS, and JavaScript, enhancing user experience and accessibility.
- Built and integrated MongoDB for efficient data storage and retrieval, improving query performance for client demo projects.

PROJECTS

Federated Learning with CNN – Mobile Health Dataset (Kaggle)

- Implemented a federated learning framework using CNNs to analyze the Mobile Health dataset from Kaggle, enabling decentralized model training across virtual clients without sharing raw data. Simulated real-world privacy-preserving training environments using PyTorch and PySyft, focusing on health data prediction.
- Conducted data preprocessing, model evaluation, and accuracy analysis across clients, demonstrating effective local training and global model aggregation.
- Tech Stack: Python, PyTorch, PySyft, Federated Averaging, CNN
- Dataset: [Mobile Health \(Kaggle\)](#)

VisnetAI – Company's Portfolio

- Developed the official website for **Visnet AI Technology Pvt Ltd**, a startup specializing in AI-driven solutions for drone surveillance and management. Utilizing **Vue.js**, I created a modular, serverless, and multi-compatible frontend framework, ensuring seamless deployment and scalability. The website effectively showcases Visnet AI's expertise in deep vision systems, including facial feature recognition, autonomous drone inspections, and advanced image and video analysis. This project highlights my proficiency in modern web development and my ability to deliver dynamic, user-centric applications aligned with strategic business objectives.
- **Live app:** <https://visnetai.co/>

Twitter Real-time Tweet Streamer

- Developed a real-time tweet streaming application utilizing MERN stack and WebSockets. The system connects to the Twitter API v2, capturing live tweets based on specified keywords and hashtags. An interactive frontend dashboard, built with React, displays these tweets in real-time, providing users with immediate insights into trending topics. The application is optimized for low latency, ensuring timely data delivery even during high traffic periods.
- **Github:** <https://github.com/Rupeshdevasam/twitter-realtime-tweet-streamer>

Certifications & Recognitions

- "Champion of the Sprint" - Awarded by Centennial Technologies for top performance and key contributions during Summer Internship.
- CSI Hackathon Winner
- Star Award - Recognized by ValueLabs for outstanding contributions.
- Certificate of Appreciation - AITAM college, for developing the official E-Learning portal.
- Certified in Python - IIT Madras, certified in C++ – IIT Bombay, certified in C – IIT Kharagpur.